State Estimation and Filtering for Robotic Perception

AA273

Conditional Expectation, Jointly Gaussian RVs, Gaussian Estimation, Linear-Gaussian SS model, Kalman Filter Equations



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 2 due Sun Jan 25
- HW 3 out Fri Jan 23, due Fri Jan 30
- Group signup sheet posted (groups from 1 – 4 people)
- Midterm: Feb 5-7
  - 5 hour timed take home exam, you choose any 5 hours in 48 hour window
  - Open book, notes, lecture videos
  - No internet, gen AI, or collaboration, coding assistants are allowed.

Lecture 6:

- Conditional Expectation
- Jointly Gaussian RVs
- Gaussian Estimation
- Linear-Gaussian SS model
- Kalman Filter Equations

Conditional Expectation

Given RV: $(X, Y) \sim p(x, y)$

Recall, cond. dist:

Conditional Expectation: $E[X \mid Y = y] = \int_x x p(x \mid y)\, dx$

$$p(x \mid y) := \frac{p(x, y)}{p(y)} = \frac{p(x, y)}{\int_x p(x, y)\, dx}$$

Notation: $\mu_{x|y} := E[X \mid Y = y]$

Conditional Expectation of a general function of X: $E[f(X) \mid Y = y] = \int_x f(x) p(x \mid y)\, dx$

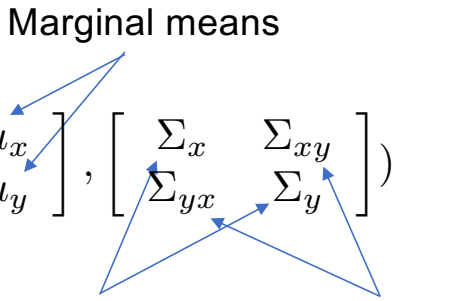
E.g. conditional covariance:

$$Cov(X \mid Y = y) = E[(X - \mu_{x|y})(X - \mu_{x|y})^T \mid Y = y]$$
$$= \int_x (x - \mu_{x|y})(x - \mu_{x|y})^T p(x \mid y)\, dx$$

Notation: $\Sigma_{x|y} := Cov(X \mid Y = y)$

Can also define conditional higher order moments, as before.

Jointly Gaussian RVs



Def.: Two joint RVs (X, Y) such that $\begin{bmatrix} X \\ Y \end{bmatrix} = Z \sim \mathcal{N}(\mu_z, \Sigma_z) = \mathcal{N}(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix})$

Notice, joint mean is the stacked vector of marginal means:

$$\mu_z = E\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} E[X] \\ E[Y] \end{bmatrix} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$$

Similarly, covariance is the block matrix of marginal and cross covariances:

$$\Sigma_z = E\left[\left(\begin{bmatrix} X \\ Y \end{bmatrix} - \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}\right)\left(\begin{bmatrix} X \\ Y \end{bmatrix} - \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}\right)^T\right]$$

$$= E\left[\begin{bmatrix} (X-\mu_x) \\ (Y-\mu_y) \end{bmatrix}\begin{bmatrix} (X-\mu_x) \\ (Y-\mu_y) \end{bmatrix}^T\right] = E\begin{bmatrix} (X-\mu_x)(X-\mu_x)^T & (X-\mu_x)(Y-\mu_y)^T \\ (Y-\mu_y)(X-\mu_x)^T & (Y-\mu_y)(Y-\mu_y)^T \end{bmatrix}$$

$$= \begin{bmatrix} E[(X-\mu_x)(X-\mu_x)^T] & E[(X-\mu_x)(Y-\mu_y)^T] \\ E[(Y-\mu_y)(X-\mu_x)^T] & E[(Y-\mu_y)(Y-\mu_y)^T] \end{bmatrix} = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}$$

Conditionals for Jointly Gaussian RVs

Claim: conditional distributions $p(x \mid y)$, $p(y \mid x)$ are also Gaussian

$$p(x \mid y) \sim \mathcal{N}(\mu_{x|y}, \Sigma_{x|y}) \qquad p(y \mid x) \sim \mathcal{N}(\mu_{y|x}, \Sigma_{y|x})$$

Equivalent Claim: for Z Gaussian, conditional dist. of any subset of elements Z_S conditioned on the other elements Z_{-S} is still Gaussian.

$$p(Z_S \mid Z_{-S}) \sim \mathcal{N}(\mu_{z_S|z_{-S}}, \Sigma_{z_S|z_{-S}})$$

where $S \subset \{1, \ldots, n\}, \quad -S = S^C$

Main points: (i) Jointly Gaussian RVs have **Gaussian marginals and Gaussian conditionals**. Gaussianity "closed" under arbitrary marginalizing and conditioning.

Main point: (ii) The posterior under Bayes' rule for jointly gaussian RVs must be Gaussian!

Gaussian Estimation Equations

How to compute posterior for jointly Gaussian RVs given their joint?

Recall: $p(x \mid y) = \dfrac{p(x,y)}{p(y)}$

And recall:

$$p(y \mid x) = \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$$

$$p(x,y) = \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}\right)$$

$$p(y) = \mathcal{N}(\mu_y, \Sigma_y)$$

Gaussian estimation equations:

$$\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y)$$

$$\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$$

Conditional mean and covariance expressed and algebraic function of joint and marginal means and covariances. **No need for integrals!**

Proof of Gaussian Estimation Equations

We will show: $p(x, y) = p(x \mid y)p(y)$

$$\eta_{xy} e^{-\frac{1}{2}\begin{bmatrix} (x-\mu_x) \\ (y-\mu_y) \end{bmatrix}^T \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}^{-1} \begin{bmatrix} (x-\mu_x) \\ (y-\mu_y) \end{bmatrix}} = \eta_{x|y} e^{-\frac{1}{2}(x-\mu_{x|y})^T \Sigma_{x|y}^{-1}(x-\mu_{x|y})} \eta_y e^{-\frac{1}{2}(y-\mu_y)^T \Sigma_y^{-1}(y-\mu_y)}$$

$$\begin{bmatrix} (x-\mu_x) \\ (y-\mu_y) \end{bmatrix}^T \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}^{-1} \begin{bmatrix} (x-\mu_x) \\ (y-\mu_y) \end{bmatrix} = (x-\mu_{x|y})^T \Sigma_{x|y}^{-1}(x-\mu_{x|y}) + (y-\mu_y)^T \Sigma_y^{-1}(y-\mu_y)$$

$$= \begin{bmatrix} (x-\mu_{x|y}) \\ (y-\mu_y) \end{bmatrix}^T \begin{bmatrix} \Sigma_{x|y} & 0 \\ 0 & \Sigma_y \end{bmatrix}^{-1} \begin{bmatrix} (x-\mu_{x|y}) \\ (y-\mu_y) \end{bmatrix}$$

Plug in Gaussian Est. Eqns:

$$\begin{cases} \mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y-\mu_y) \\ \Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \end{cases}$$

On the board and in the lecture notes. Application of Woodbury Matrix inverse Lemma

Linear-Gaussian States Space Model

Dynamics eqn.: $X_{t+1} = A_t X_t + B_t u_t + W_t \qquad W_t \sim GWN(0, Q_t)$

Measurement eqn.: $Y_t = C_t X_t + D_t u_t + V_t \qquad V_t \sim GWN(0, R_t)$

Initial condition: $X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$ $\qquad X_0, W_t, V_\tau$ independent for all t, τ

Claim: posterior $p(x_t \mid y_{1:t})$ and prediction $p(x_{t+1} \mid y_{1:t})$ is Gaussian for all t.

Notation: $p(x_t \mid y_{1:t}) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})$

$$p(x_{t+1} \mid y_{1:t}) = \mathcal{N}(\mu_{t+1|t}, \Sigma_{t+1|t})$$

We compute these with the Kalman Filter

Kalman Filter Equations

Given: $\mu_{0|0}, \Sigma_{0|0},\ A_t, B_t, C_t,\ Q_t, R_t$

(Assume $D_t = 0$)

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_tC_t\Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

Compare update with Gaussian Est. Eqns:

$$\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y)$$

$$\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$$

Notice the same form!

We'll derive this next time.